TOTALITY

A generator of hypnotic Berlin techno

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About

Totality composes, mixes and plays hypnotic Berlin techno: tracks of six to eight minutes and DJ sets of up to four hours, in which the tracks follow one another on two decks as a DJ would mix them. Nothing is played back from a recording. The kick, the rumble under it, the twelve lanes of the drum kit, the ping, the bass and the 303, the dub chord, the drone and the texture are synthesised; the tape echo, the spring, the plate and the rooms are simulated; and every note, every operation of the form and every move of a knob is decided by the composer.

The music it aims at is the one of Berghain and Ostgut Ton, of Dub Techno and of the hypnotic school: a groove that holds for minutes and changes by one element at a time, a low end made of a kick and the rumble of its own reverb, a chord thrown into a tape echo. Four style profiles span it -- Hypnotic in the centre, Ostgut, Dub and Raw -- and every knob of their recipes was fitted against measurements of thirty reference tracks: their spectral balance, their width, their loudness and its range, and how alike their bars are.

The same seed always gives the same track. A track is a seed, a style, a length and a handful of rerolls, which is what a set file stores and what you can hand to someone else. The standalone and the plugin play what the offline renderer plays, sample for sample.

Totality is the moment of a total eclipse when the moon covers the sun entirely and only the corona shines round the dark disc. In Totality's Patterns view the kick is that disc: every other part turns around it in rings, and its corona swells with every beat.

Quick start

The standalone plays by itself: press Play. Track and DJ mix, beside the style, choose what Totality makes: a single track, or a DJ mix -- a set of tracks mixed on two decks. A click on either composes it at once; the length beside them is the one of what is chosen (a track 2 to 16 minutes, a mix 10 minutes to 12 hours), and the Compose button names what it writes (Compose track, Compose mix) with the knobs as they stand; the die draws a fresh seed. The status line begins with "single track" or "DJ mix of" and tells what plays: its style and form, its tempo, its key with the Camelot label a DJ mixes by, and its length. While a track begins to play, Totality measures its loudest part in the background, sets every part against the kick -- each lane of the kit, the figure and the other tonal voices, one or two up front and the rest behind them, the room taken back where the mids would stand over the records' -- and brings the track to its style's loudness; the status says "levelling" until the corrections have glided in.

A DJ mix of that many minutes is a set: a dramaturgy of tempo and energy, tracks in neighbouring keys, blends of 16 or 32 bars with the bass swapped on the 32-bar line where the incoming track's figure lands -- its first body block runs under the outgoing track's full groove --, loops of the last track under the next on the third deck, breaks from the mixer's effects.

In a DAW, Totality is an instrument plugin that follows the host's transport: it plays when the host plays, jumps when the host jumps, and plays at the host's tempo (a set's own tempo drift is the standalone's and the export's). Its output is a finished stereo mix at about -10 LUFS with its true peak under -1 dBTP.

A reroll draws one part of the track again and leaves the rest as it was: form, harmony, rack, layers, blocks, events, moves, sounds. In a mix the rerolls act on the track under the playhead, and "reroll the whole track" or "the mix's plan" draw more at once (the Arrange page).

The panel

Totality shares its panel with the other generators of the family -- Phosphene, Ephemeris and Parhelion (and, in part, Noctuary) -- so that what is learned on one holds on the others: the same places, the same keys, the same controllers, the same hands. Each keeps its own colours, its own letters and the picture behind its panel (Settings > Picture behind the panel takes it away).

The header's first row: the logo, the style, the key and the scale, what is composed and its length, Compose, the die (a new seed), Play and the speaker (mute). Its second row: the thumbs -- a verdict on what plays, for the curation --, the status line, and at the right Undo, Redo, Help and the settings. Under the header the whole track or set at a glance; a click jumps there. The tabs come in one order everywhere: Set, Arrange, the instrument's own pages -- a page of several has small tabs of its own --, Mixer, Perform, Export, Style.

The unambiguous actions are icons since 03.10.2026, their words in the tooltips: the die draws a new seed, the triangle plays and the square stops, the speaker mutes (crossed out while muted), the disk saves a set and the folder loads one (Export).

The ways around the panel (15.10.2026, the family's). The magnifier at the right of the header (Ctrl+K) opens the search: a few letters find any knob -- its page opens, its section or lane comes to the front and it lights up; a number after its name sets it ("cutoff 800") --, any preset of any synth, a page or an action. The line along the bottom of the window says what the control under the mouse is: its value now, its range and default, where the score plays it, what moves it, its MIDI controller and its key. Each synth's presets have a magnifier of their own, the browser: by name, by group, the starred and the recently played; hold a row's triangle to hear it, double-click to take it. The disk beside saves the sound as a preset of your own (Documents/Totality/Presets), listed under Yours, and a star says that the knobs have moved from the preset. The first steps show the panel on the first start (Settings > First steps, again). The standalone's window comes back where it was left -- its place, its size, maximised or not --, and its audio settings offer ASIO (Windows).

A page taller than the window shows its groups in sections, one at a time, with a switch at its top right: a synth's sound and its modulation apart (Sound, Modulation), and where even those do not fit, cut further, each named after its first group (Filter, LFO, Matrix ...). At the window's usual size, 1280 by 860, no page scrolls. The overview above the tabs can be folded away in the settings; the pages then get its height.

A colour says what a control does, the same on every page and in every generator of the family: the sources, the filters, the envelopes, what moves by itself, the room and the mix. A knob the composer moves has a thin ring round it that shows where it stands at this moment. A double click sets a control to its default; a right click offers MIDI learn, forgetting the controller, and the default.

Undo and Redo take back and bring back what was changed -- a knob, a new seed, a reroll, a preset, a loaded set -- one step at a time; the button's tooltip names the step. Help opens this manual by topic, with the page in front, the keys and the headset as topics of their own. The settings hold the update check (it asks GitHub once a day whether a newer Totality is out; nothing else is sent), the picture behind the panel, the overview, the headset, the window's size, full screen, the keys and About.

```
Space                play / stop
Ctrl+Z               undo
Ctrl+Y, Ctrl+Shift+Z  redo
F1                   help
F11                  full screen (the standalone); Esc leaves it and closes the help
Ctrl+S, Ctrl+O       save the set, open one
```

```
Ctrl+E                export
Ctrl+K                search: any control, preset, page or action
```

The controllers every generator of the family listens to without learning: controller 74 (brightness) moves the master filter, 11 (the expression pedal) the echo throw; the mod wheel is left free. The keys from middle C up toggle the mutes (Perform).

The headset. The controls of a Meta Quest's hands are shown only while a headset sends them -- or always, if the settings say so (Headset > Always show them); Off stops listening. The Quest app sends its hands to this computer in bridge mode: bridge_host = this computer's address in its tot.cfg (audio=0 leaves the headset silent), and Totality listens on port 9103. The hands mean the same in every generator of the family, on the headset alone and here:

```
left pinch            play / stop
both hands pinched    the next track (or set)
right pinch           kick out / kick in
left hand up, down    the master filter: low pass below the middle, high pass above
right hand up         the echo throw
```

A pinch acts when it opens again, so a pinch of both hands is never also one of each; a hand moves its control only while it does not pinch, and both glide and are centred: a hand at mid height plays what was composed.

The track and the set



A track has one of three forms. The Arc is the DJ tool: an intro of 32 or 64 bars -- never the kick alone: kick, hats and a percussion from the first bar, or hats and percussion, or an atmosphere and percussion, with the kick on bar 9 or 17, as most of the reference records begin (the plan names the opening); the percussion that opens it carries it, four to seven hits a bar and up front in the balance, and a mix's first track never begins with the kick --, a body that builds its groove over its first blocks -- a main entry on each block's first bar and further hats and percussion on its 8-bar lines, full by about bar 97 --, an outro that takes them away again -- so it mixes. The Peak builds to a kick-out and returns into its densest block. The Endless is full from the first bar and changes only by exchanging layers, as the hypnotic records do. Every block has exactly one operation (add, remove, swap) on its first bar -- a hat or a perc may come in on the block's 8- or 16-bar line instead --, every change falls on a 4-bar line and a return on a 16-bar line; nothing tonal plays in the first and last 32 bars, so two tracks can be mixed without a clash.

Every track is of a kind as well as of a style, its archetype: a Tool (the groove and its automation), a Roller (toms, shaker and rim in polymeters, the percussion up front), a Stab, an Acid or a Bleep track (the chord's stabs, a 303 line or the ping as its figure), a Dub Chord track (the chord with its echo chain) or a Tribal track (toms, claps and rims, fills, a kick-out). Each style draws them with its own weights -- Hypnotic mostly Rollers and Bleeps, Ostgut Stabs and Tools, Dub its Dub Chords, Raw Acid and Tribal tracks --; Archetype fixes one, and a set never plays two of a kind in a row. The archetypes with a melodic figure play it the same one or two bars from its entry (the body's second block; a bass riff with the body) to the end, and it changes only through its

filters. Between the operations the body runs in waves of 32 or 64 bars: the figure's filters and the hats' bus open towards a wave's end and fall back after it; two to four bars before that landing something drops away -- the centre of the mix, the kick and the claps, the figure -- so that the landing lands; in the middle of a wave one element (the figure, the bass or the rumble, the percussion, the open hats and the ride) breathes out for 8 or 16 bars and comes back with an echo throw. The Arrange page names the figure and the bars where the waves land.

Style picks the profile; Morph To and Morph blend it with a second one; the Dub and Hypnotic shares pull any profile along the two axes the style is described by. With Auto the tempo, the key and the owner of the low end are drawn from the profile; without it the knobs decide. Minutes is the track's length, rounded to whole blocks.

A set is mixed the way the Berghain residents describe their nights: as one long recomposition, not a row of crossfades. Each track is heard for about three minutes of its own (Track Time) and a third of it lies under its neighbours -- about twenty tracks an hour. The incoming track comes in with its first body block, its highs first and its mids over the last 16 bars, its low band killed; on a 32-bar line the bass goes over in one move; the outgoing track keeps its hats a while, loses its mids and highs and fades. The third deck borrows: the hats and percussion of the track that just left carry on under the new one, the next track's figure is teased in, high-passed, before its blend (where the keys agree), or a percussion loop of the track before last is layered under the one playing. Between the blends the DJ's hand works the channel on the 16-bar lines -- a low kill before the line so the kick slams back, the highs taken out and faded back in, a dip of the mids, the high-pass filter drawn up into the line, an echo throw on the last beat -- never where the track itself already drops something.

The set knobs: its length (up to twelve hours), the dramaturgy (Warm-up, Peak, Closing, Sunday, Flat, Cruise -- hard from the start, a long cruise, building again at the end -- and Marathon -- a warm-up, two peaks, a long closing: the arc of tempo and energy over its length), whether the styles wander with the energy or stay, how often the third deck borrows (Live Loops), how often a break comes from the mixer's effects, the blend's length, a track's own time (Track Time) and how busy the DJ's hand is (DJ Hand). The mixer's effects are one send per deck into a tape echo and a long hall, which the set's breaks throw into and the Perform page's throw uses.

COMPOSE

Control	Key	Range	Default
Tempo	`compose.bpm`	100 .. 160 BPM	130 BPM
Key	`compose.key`	C, C#, D, D#, E, F, F#, G, G#, A, A#, B	A
Scale	`compose.scale`	Aeolian, Dorian, Phrygian, Hexachord, Minor Pentatonic	Aeolian
Style	`compose.style`	Hypnotic, Ostgut, Dub, Raw Peak	Hypnotic
Length	`compose.minutes`	2 .. 16 min	7 min
Swing	`compose.swing`	50 .. 66 %	53 %
Humanize	`compose.humanize`	0 .. 10 ms	3 ms
Low End	`compose.low_owner`	Rumble, Sub	Rumble
Auto	`compose.auto`	off / on	on
Form	`compose.form`	Auto, Arc, Peak, Endless	Auto
Morph To	`compose.morph_to`	Off, Hypnotic, Ostgut, Dub, Raw Peak	Off
Morph	`compose.morph`	0 .. 1	0.5

Dub Share	`compose.dub_share`	0 .. 1	0
Hypnotic Share	`compose.hypnotic_share`	0 .. 1	0
Composer's Sounds	`compose.pick_sounds`	off / on	on
Archetype	`compose.archetype`	Auto, Tool, Roller, Stab, Acid, Bleep, Dub Chord, Tribal	Auto
Favor Ratings	`compose.use_ratings`	off / on	off

SET

Control	Key	Range	Default
Dramaturgy	`set.dramaturgy`	Warm-up, Peak, Closing, Sunday, Flat, Cruise, Marathon	Peak
Styles	`set.journey`	Stay, Wander	Wander
Live Loops	`set.loops`	0 .. 1	0.5
FX Breaks	`set.fx_breaks`	0 .. 1	0.3
Blend	`set.blend`	16 bars, 32 bars	32 bars
Set Length	`set.minutes`	0 .. 720 min	0 min
Track Time	`set.track_minutes`	2 .. 6 min	3.5 min
DJ Moves	`set.dj_hand`	0 .. 1	0.5

DJFX

Control	Key	Range	Default
Echo Time	`djfx.echo_time`	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/16
Feedback	`djfx.feedback`	0 .. 1	0.45
Echo Return	`djfx.echo_return`	-60 .. 6 dB	-3 dB
Hall Decay	`djfx.hall_decay`	0.5 .. 12 s	5 s
Hall Return	`djfx.hall_return`	-60 .. 12 dB	6 dB

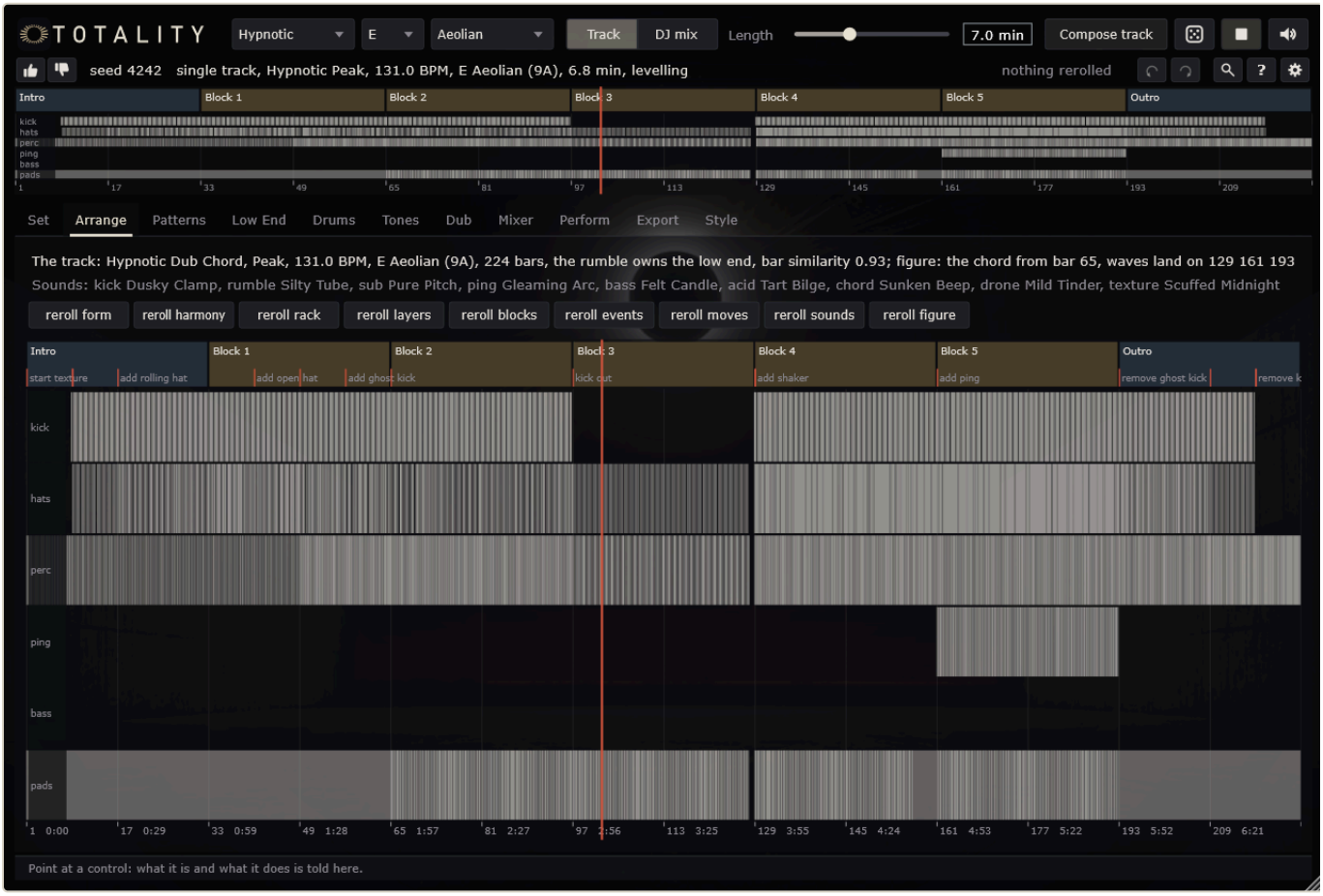
Sounds and presets

Every synth has 1024 factory presets: the kick, the rumble, the sub, a lane of the kit, the ping, the bass, the 303, the dub chord, the drone and the texture. Each engine has sixteen groups -- "Berlin 909", "Dub Sine", "Ride Bell", "Classic 303", "Basic Channel" -- and each group sixty-four presets on an eight by eight grid: the rows an adjective from dark to bright ("Muffled" ... "Cutting"), the columns a noun of the group's own, so the name tells where on the grid a sound lies. A preset is the sound, not the mix: levels, pans, sends, ducking, the octaves and a lane's role and pattern stay where the mix and the composer put them.

For every track the composer chooses a preset per synth and per lane of the kit (Composer's Sounds, compose.pick_sounds): a group by how well it suits the track's style, a lane only among the presets made for its role, and in the group a row that leans to the middle -- the darkest and the brightest are chosen less often. The style then holds a few knobs in its own range (the Dub kick's soft click, the Raw kick's hard clip, the Dub chord's bright band). Reroll sounds draws other presets and leaves the notes as they are.

The knobs show what plays. When a track begins -- in a set, when the next track starts on its deck -- its presets and its style's mix are put on the knobs, and each page names the preset the composer chose ("this track: Punchy Drum Hall (Hypnotic Thud)"); the Arrange page lists them for the whole track. Turn a knob and the sound follows from there; choose another preset from the menu, or step through them, and it plays at once. In a blend each deck keeps its own track's sound: the knobs show the incoming track while the outgoing one plays on as it was.

Arrange



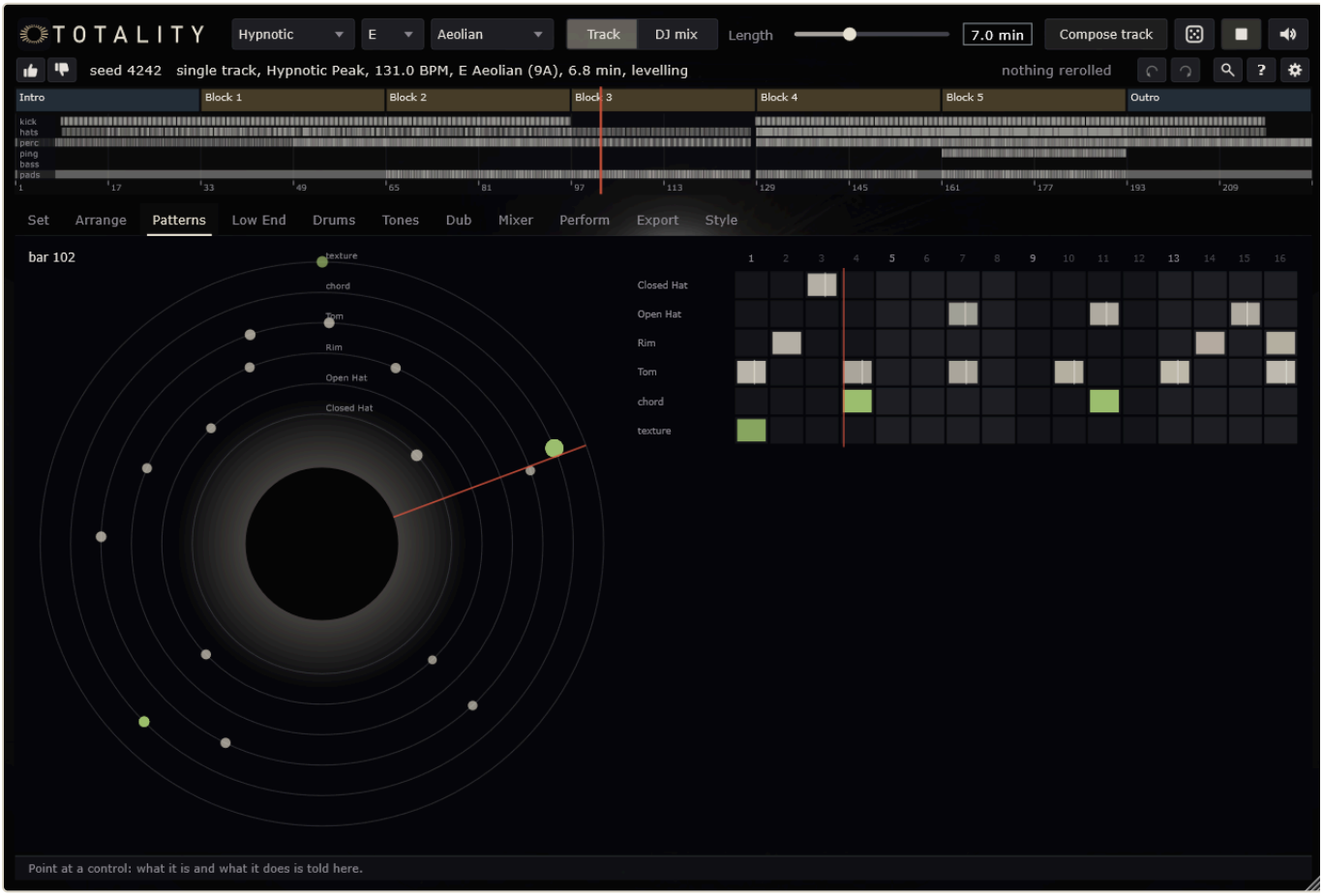
The arrangement across the whole length: in a mix the tracks on decks A and B (their swaps marked: from there a track owns the low end), in a track its blocks with their names, and under them a lane per group of layers -- kick, hats, perc, ping, bass, pads -- lit where it plays, and a ruler: a track's bars (on the Arrange page with the time beside them), a mix's minutes. The Arrange page shows the operations of the form by name. A click jumps there.

The mouse wheel zooms in and out around the pointer, down to four bars, where every hit of the kit stands apart; a drag, the wheel with Shift or a sideways wheel moves along, a double click shows the whole length again. A zoomed view pages on with the playhead when it runs out of it (not when you moved the view away), and the strip on top marks the window of the Arrange page.

The row of rerolls acts on the track under the playhead; each part of it is drawn from a stream of its own, so a reroll of the blocks keeps the harmony and the sounds, a reroll of the moves (the automation) keeps every note, and a reroll of the figure draws another motif for the same voice.

The + and - beside the status line rate the track under the playhead. The ratings are kept in Documents/Totality/ratings.tsv with the track's archetype and the groups of its presets; with Favor Ratings (the Set page) the composer draws the archetypes and preset groups you liked more often and the ones you did not seldom -- none is ever ruled out. A saved set does not carry the ratings: it plays back the same only with Favor Ratings off.

Patterns and the Eclipse



The kick is the dark disc in the middle, and every other part that plays in the bar is a ring around it: a bar once round, its onsets as beads, larger the louder, the playhead's hand sweeping over them. A layer on the sixteen-step matrix shows the same beads bar after bar; a polymeter or a slipping layer moves on from bar to bar -- it precesses, and you can see it. Where three rings or more have an onset on the same sixteenth, the corona lights up at that angle: a conjunction. Beside it the same bar as a step grid, with a tick where an onset lies off the grid (swing and feel).

Low End


TOTALITY
Hypnotic
E
Aeolian
Track
DJ mix
Length
7.0 min
Compose track
seed 4242 single track, Hypnotic Peak, 131.0 BPM, E Aeolian (9A), 6.8 min, levelling
nothing rerolled
Intro
Block 1
Block 2
Block 3
Block 4
Block 5
Outro
Set
Arrange
Patterns
Low End
Drums
Tones
Dub
Mixer
Perform
Export
Style
Kick
Rumble
Sub
KICK PRESET
Dusky Clamp
this track: Dusky Clamp (Tight Tool)
ENGINE
Engine
Tune
Level
Drive
Clip
909
Key
Soft
-3.00 dB
0.47
PITCH
Start
End
Body Decay
300 Hz
52.0 Hz
15.5 ms
PUNCH AND AMP
Punch
Punch Decay
Attack
Hold
Decay
Tail Limit
0.58
3.00 ms
0.20 ms
25.0 ms
300 ms
0.00 dB
CLICK
Click
Tone
Decay
0.35
3000 Hz
6.00 ms
TONE
Tone
Low Cut
Dip Freq
Dip
6080 Hz
30.0 Hz
500 Hz
-3.00 dB
TOP LAYER
Top Level
Top Pitch
Top Decay
Top Drive
Top Cut
-12.0 dB
-8.00 st
90.0 ms
0.60
400 Hz
Point at a control: what it is and what it does is told here.



Low End, its second small tab: the rumble.



Low End, its third small tab: the sub.

The low end is one system. The kick hands its phase on: the rumble's sub continues the kick's tail at the phase the kick left it, and the sub bass locks its phase to the last kick, so the three never cancel each other. The kick has three engines (a swept sine, a resonator, a modelled 909) and a top layer; its drive runs at four times the rate.

The rumble is the kick sent into a hall, cut below the split, clipped and ducked by the next kick -- the dark roll of Berlin techno. Either the rumble or the sub bass owns the band under 80 Hz (compose.low_owner): never both at once.

KICK

Control	Key	Range	Default
Engine	`kick.engine`	Sweep, Resonator, 909	Sweep
Tune	`kick.tune`	Free, Key, Fifth, Flat Seventh	Key
Pitch End	`kick.pitch_end`	30 .. 120 Hz	52 Hz
Pitch Start	`kick.pitch_start`	60 .. 1500 Hz	300 Hz
Body Decay	`kick.pitch_decay`	5 .. 300 ms	18 ms
Punch Decay	`kick.punch_decay`	0.5 .. 20 ms	3 ms
Punch	`kick.punch`	0 .. 1	0.45

Attack	`kick.amp_attack`	0 .. 10 ms	0.2 ms
Hold	`kick.amp_hold`	0 .. 150 ms	25 ms
Decay	`kick.amp_decay`	20 .. 1500 ms	380 ms
Drive	`kick.drive`	0 .. 1	0.35
Clip	`kick.clip`	Soft, Hard	Soft
Click	`kick.click_level`	0 .. 1	0.35
Click Tone	`kick.click_tone`	500 .. 12000 Hz	3000 Hz
Click Decay	`kick.click_decay`	0.5 .. 30 ms	6 ms
Tone	`kick.tone`	200 .. 20000 Hz	6000 Hz
Level	`kick.level`	-36 .. 6 dB	-3 dB
Tail Limit	`kick.tail_limit`	-60 .. 0 dB	0 dB
Low Cut	`kick.low_cut`	15 .. 60 Hz	30 Hz
Dip Freq	`kick.dip_freq`	200 .. 1500 Hz	500 Hz
Dip	`kick.dip`	-12 .. 0 dB	-3 dB
Top Level	`kick.top_level`	-60 .. 0 dB	-12 dB
Top Pitch	`kick.top_pitch`	-24 .. 0 st	-8 st
Top Decay	`kick.top_decay`	20 .. 400 ms	90 ms
Top Drive	`kick.top_drive`	0 .. 1	0.6
Top Cut	`kick.top_cut`	150 .. 2000 Hz	400 Hz

RUMBLE

Control	Key	Range	Default
Level	`rumble.level`	-60 .. 6 dB	-9 dB
Split	`rumble.split`	50 .. 150 Hz	80 Hz
Decay	`rumble.decay`	0.5 .. 6 s	1.8 s
Size	`rumble.size`	0.3 .. 3	1
Damping	`rumble.damping`	0 .. 1	0.5
Pre-Delay	`rumble.predelay`	0 .. 60 ms	0 ms
Drive	`rumble.drive`	0 .. 18 dB	6 dB
Low Pass	`rumble.ratio`	1.5 .. 6 x f0	3 x f0
Resonance	`rumble.resonance`	0 .. 1	0.15
Sub	`rumble.sub`	-60 .. 6 dB	0 dB
Sub Attack	`rumble.sub_attack`	5 .. 200 ms	25 ms
Sub Release	`rumble.sub_release`	50 .. 2000 ms	300 ms
Duck	`rumble.duck`	0 .. 24 dB	7 dB
Duck Hold	`rumble.duck_hold`	0 .. 250 ms	80 ms
Duck Release	`rumble.duck_release`	30 .. 600 ms	220 ms

SUB

Control	Key	Range	Default
Level	`sub.level`	-60 .. 6 dB	-6 dB
Octave	`sub.octave`	-2 .. 2	0
Attack	`sub.attack`	0.5 .. 50 ms	3 ms
Decay	`sub.decay`	20 .. 2000 ms	250 ms
Sustain	`sub.sustain`	0 .. 1	0.7
Release	`sub.release`	10 .. 800 ms	90 ms
Low Pass	`sub.low_pass`	40 .. 300 Hz	110 Hz
Drive	`sub.drive`	0 .. 1	0.1
Kick Lock	`sub.lock`	Off, Kick	Kick
Duck	`sub.duck`	0 .. 24 dB	9 dB
Duck Hold	`sub.duck_hold`	0 .. 250 ms	40 ms
Duck Release	`sub.duck_release`	30 .. 600 ms	180 ms

Drums



Twelve lanes, each a small drum synthesiser: a role (the closed and rolling hats, the open hat, the ride, the claps, the snare, the rim, the shaker, toms and congas, noise), an engine (noise, metal -- the 909's six square waves -- modal, tone, FM), a filter, a shape. They run side by side in the lanes of the processor's vector registers. The hats and the percussion each have a bus with its level and a low pass, then the drum bus's saturation; the ducks let the pads and the rooms step aside for the kick, band by band.

PERC (12 INSTANCES)

Control	Key	Range	Default
Active	`percN.active`	off / on	on
Role	`percN.role`	Closed Hat, Rolling Hat, Open Hat, Ride, Clap, Clap Ghost, Snare, Rim, Shaker, Tom, Conga, Noise	Closed Hat
Engine	`percN.engine`	Noise, Metal, Modal, Tone, FM	Noise
Pitch	`percN.pitch`	40 .. 12000 Hz	400 Hz
Pitch Amount	`percN.pitch_amount`	1 .. 16 x	1 x
Pitch Decay	`percN.pitch_decay`	0.5 .. 300 ms	10 ms
FM Ratio	`percN.fm_ratio`	0.25 .. 8	1.41
FM Index	`percN.fm_index`	0 .. 8	0

Modes	`percN.mode_set`	Membrane, Bar, Harmonic	Membrane
Mode Damping	`percN.mode_damp`	0 .. 1	0.5
Metal Scale	`percN.metal_scale`	0.25 .. 4 x	1 x
Noise	`percN.noise`	0 .. 1	1
Noise Decay	`percN.noise_decay`	2 .. 3000 ms	70 ms
Bursts	`percN.bursts`	1 .. 6	1
Burst Spacing	`percN.burst_spacing`	2 .. 40 ms	10 ms
Decay	`percN.decay`	2 .. 3000 ms	70 ms
Filter	`percN.filter`	Low Pass, Band Pass, High Pass	High Pass
Cutoff	`percN.cutoff`	100 .. 18000 Hz	7000 Hz
Resonance	`percN.resonance`	0 .. 1	0.15
Low Cut	`percN.low_cut`	150 .. 8000 Hz	3000 Hz
Drive	`percN.drive`	0 .. 1	0.1
Level	`percN.level`	-36 .. 6 dB	-0.7 dB
Pan	`percN.pan`	-1 .. 1	0.1
Choke Group	`percN.choke`	0 .. 4	1
Shift	`percN.shift`	-10 .. 10 ms	0 ms
Density	`percN.density`	0 .. 1	0.5
Tune to Key	`percN.tune`	off / on	off
Pan Depth	`percN.pan_depth`	0 .. 1	0.3
Pan Period	`percN.pan_bars`	0.0625 .. 16 bars	0.1875 bars
Cut Tracks Pitch	`percN.cut_track`	0 .. 2	0
Noise Type	`percN.noise_type`	White, 909 Metal	909 Metal

MIX

Control	Key	Range	Default
Hats Level	`mix.hats_level`	-24 .. 12 dB	-3.5 dB
Hats Cut	`mix.hats_cut`	200 .. 20000 Hz	20000 Hz
Perc Level	`mix.perc_level`	-24 .. 12 dB	-6.5 dB
Perc Cut	`mix.perc_cut`	200 .. 20000 Hz	20000 Hz
Drum Saturation	`mix.drum_sat`	0 .. 1	0.35
Duck Low	`mix.duck_low`	0 .. 18 dB	10 dB
Duck Mid	`mix.duck_mid`	0 .. 12 dB	3 dB
Group Low Cut	`mix.low_cut`	20 .. 500 Hz	20 Hz

Tones


TOTALITY
Hypnotic
E
Aeolian
Track
DJ mix
Length
7.0 min
Compose track
seed 4242 single track, Hypnotic Peak, 131.0 BPM, E Aeolian (9A), 6.8 min, levelling
nothing rerolled
Intro
Block 1
Block 2
Block 3
Block 4
Block 5
Outro
kick
hats
perc
ping
bass
pads
Set
Arrange
Patterns
Low End
Drums
Tones
Dub
Mixer
Perform
Export
Style
Ping
Bass
Acid
Chord
Drone
Texture
PING PRESET
Gleaming Arc
this track: Gleaming Arc (Sweeping Ping)
VOICE
Level
Pan
Width
Ratio
Index
Index Decay
Pitch Amount
Pitch Decay
-15.5 dB
0.00
0.50
1.88
2.00
60.0 ms
0.80
45.0 ms
GATE
Decay
Low-Pass Gate
Release
Resonance
114 ms
0.60
150 ms
0.35
BAND
Band
Q
Mix
Sweep
Sweep Rate
1406 Hz
1.21
0.50
1.28 oct
0.06 Hz
Point at a control: what it is and what it does is told here.



Tones: the bass.



Tones: the acid.



Tones: the chord.



Tones: the drone.



Tones: the texture.

The ping is the hypnotic school's figure: FM through a low-pass gate, a band pass that wanders. The bass and the 303 share one synthesiser with ten circuit filters (the Moog ladder, the 303's diode ladder, the Prophet's SSM2040, the Juno's IR3109, the SEM, the MS-20's Korg35 and more); the 303 line has its accents and slides. The dub chord is a minor seventh or a stab of six voices through a band pass, a crusher and a phaser, sent into the tape echo and the plate. The drone holds a note under eight bars; the texture is vinyl crackle, hum and erosion.

PING

Control	Key	Range	Default
Level	`ping.level`	-60 .. 6 dB	-12.5 dB
Pan	`ping.pan`	-1 .. 1	0
Width	`ping.width`	0 .. 1	0.5
Ratio	`ping.ratio`	0.25 .. 8	1.414
Index	`ping.index`	0 .. 6	2
Index Decay	`ping.index_decay`	2 .. 500 ms	60 ms
Pitch Amount	`ping.pitch_amount`	0 .. 4	0.8
Pitch Decay	`ping.pitch_decay`	2 .. 300 ms	45 ms
Decay	`ping.decay`	10 .. 2000 ms	140 ms
Low-Pass Gate	`ping.lpg`	0 .. 1	0.6
Gate Release	`ping.lpg_release`	10 .. 2000 ms	150 ms

Resonance	`ping.resonance`	0 .. 1	0.35
Band	`ping.band`	150 .. 6000 Hz	900 Hz
Band Q	`ping.band_q`	0.2 .. 8	0.8
Band Mix	`ping.band_mix`	0 .. 1	0.5
Sweep	`ping.sweep`	0 .. 2 oct	0.6 oct
Sweep Rate	`ping.sweep_rate`	0.01 .. 2 Hz	0.07 Hz

BASS

Control	Key	Range	Default
Level	`bass.level`	-60 .. 6 dB	-9 dB
Pan	`bass.pan`	-1 .. 1	0
Wave	`bass.wave`	0 .. 1	0
Pulse Width	`bass.pulse_width`	0.05 .. 0.95	0.5
Sub Osc	`bass.sub_osc`	0 .. 1	0.3
Filter	`bass.filter`	Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder (303), Korg35 (MS-20), Polivoks, EDP Wasp, Comb	Juno (IR3109)
Cutoff	`bass.cutoff`	40 .. 12000 Hz	350 Hz
Resonance	`bass.resonance`	0 .. 1	0.2
Env Amount	`bass.env_amount`	0 .. 6 oct	2 oct
Decay	`bass.decay`	5 .. 3000 ms	180 ms
Accent	`bass.accent`	0 .. 1	0.3
Attack	`bass.amp_attack`	0.5 .. 500 ms	3 ms
Amp Decay	`bass.amp_decay`	5 .. 3000 ms	250 ms
Sustain	`bass.amp_sustain`	0 .. 1	0.6
Release	`bass.amp_release`	5 .. 3000 ms	300 ms
Glide	`bass.glide`	0 .. 500 ms	0 ms
Drive	`bass.drive`	0 .. 1	0.2
Key Track	`bass.key_track`	0 .. 1	0.5
Low Cut	`bass.low_cut`	20 .. 500 Hz	65 Hz
High Cut	`bass.high_cut`	500 .. 20000 Hz	5000 Hz
Dub Send	`bass.dub_send`	0 .. 1	0
Room Send	`bass.room_send`	0 .. 1	0
Duck	`bass.duck`	0 .. 24 dB	8 dB
Duck Release	`bass.duck_release`	30 .. 600 ms	200 ms

ACID

Control	Key	Range	Default
Level	`acid.level`	-60 .. 6 dB	-14 dB

Pan	`acid.pan`	-1 .. 1	0.1
Wave	`acid.wave`	0 .. 1	0
Pulse Width	`acid.pulse_width`	0.05 .. 0.95	0.5
Sub Osc	`acid.sub_osc`	0 .. 1	0
Filter	`acid.filter`	Moog Ladder, Prophet (SSM2040), Juno (IR3109), Oberheim SEM, Xpander, Diode Ladder (303), Korg35 (MS-20), Polivoks, EDP Wasp, Comb	Diode Ladder (303)
Cutoff	`acid.cutoff`	40 .. 12000 Hz	600 Hz
Resonance	`acid.resonance`	0 .. 1	0.72
Env Amount	`acid.env_amount`	0 .. 6 oct	2.5 oct
Decay	`acid.decay`	5 .. 3000 ms	220 ms
Accent	`acid.accent`	0 .. 1	0.6
Attack	`acid.amp_attack`	0.5 .. 500 ms	3 ms
Amp Decay	`acid.amp_decay`	5 .. 3000 ms	250 ms
Sustain	`acid.amp_sustain`	0 .. 1	0.8
Release	`acid.amp_release`	5 .. 3000 ms	60 ms
Glide	`acid.glide`	0 .. 500 ms	60 ms
Drive	`acid.drive`	0 .. 1	0.5
Key Track	`acid.key_track`	0 .. 1	0.3
Low Cut	`acid.low_cut`	20 .. 500 Hz	120 Hz
High Cut	`acid.high_cut`	500 .. 20000 Hz	9000 Hz
Dub Send	`acid.dub_send`	0 .. 1	0.25
Room Send	`acid.room_send`	0 .. 1	0.01
Duck	`acid.duck`	0 .. 24 dB	4 dB
Duck Release	`acid.duck_release`	30 .. 600 ms	200 ms

CHORD

Control	Key	Range	Default
Level	`chord.level`	-60 .. 6 dB	-5 dB
Width	`chord.width`	0 .. 1	0.6
Detune	`chord.detune`	0 .. 30 ct	10 ct
Octave	`chord.octave`	-2 .. 2	0
Attack	`chord.attack`	0.5 .. 500 ms	10 ms
Decay	`chord.decay`	20 .. 3000 ms	300 ms
Sustain	`chord.sustain`	0 .. 1	0.15
Release	`chord.release`	10 .. 3000 ms	400 ms
Brightness	`chord.bright`	200 .. 12000 Hz	2500 Hz
Env Amount	`chord.env_amount`	0 .. 5 oct	1.5 oct
Band	`chord.band`	100 .. 3000 Hz	380 Hz

Band Mode	`chord.band_mode`	Band Pass, High Pass	Band Pass
Band Q	`chord.band_q`	0.2 .. 4	0.7
Sweep	`chord.sweep`	0 .. 2 oct	0.5 oct
Sweep Rate	`chord.sweep_rate`	0.01 .. 2 Hz	0.2 Hz
Bits	`chord.crush`	4 .. 16	8
Crush Mix	`chord.crush_mix`	0 .. 1	0.2
Phaser	`chord.phaser`	0 .. 1	0.5
Phaser Rate	`chord.phaser_rate`	0.05 .. 4 Hz	0.5 Hz
Dip 600 Hz	`chord.dip`	-18 .. 0 dB	-7 dB
Dub Send	`chord.dub_send`	0 .. 1	0.5
Plate Send	`chord.plate_send`	0 .. 1	0.35

DRONE

Control	Key	Range	Default
Level	`drone.level`	-60 .. 6 dB	-13 dB
Octave	`drone.octave`	-2 .. 2	0
Detune	`drone.detune`	0 .. 30 ct	8 ct
Cutoff	`drone.cutoff`	80 .. 8000 Hz	700 Hz
Resonance	`drone.resonance`	0 .. 1	0.25
Sweep	`drone.sweep`	0 .. 4 oct	1.5 oct
Sweep Bars	`drone.sweep_bars`	4 .. 128 bars	32 bars
Attack	`drone.attack`	0.01 .. 20 s	4 s
Release	`drone.release`	0.01 .. 20 s	6 s
Plate Send	`drone.plate_send`	0 .. 1	0.4
Room Send	`drone.room_send`	0 .. 1	0.02

TEXTURE

Control	Key	Range	Default
Level	`texture.level`	-60 .. 0 dB	-26 dB
Crackle	`texture.crackle`	0 .. 1	0.5
Hum	`texture.hum`	0 .. 1	0.2
Hum Freq	`texture.hum_hz`	40 .. 70 Hz	50 Hz
Erosion	`texture.erosion`	0 .. 1	0.4
Width	`texture.width`	0 .. 1	0.8

Dub


TOTALITY
Hypnotic
E
Aeolian
Track
DJ mix
Length
7.0 min
Compose track
seed 4242 single track, Hypnotic Peak, 131.0 BPM, E Aeolian (9A), 6.8 min, levelling
nothing rerolled
Intro
Block 1
Block 2
Block 3
Block 4
Block 5
Outro
kick
hats
perc
ping
bass
pads
Set
Arrange
Patterns
Low End
Drums
Tones
Dub
Mixer
Perform
Export
Style
Dub
Space
Cloud
TAPE ECHO
Echo Time
1/4
Feedback
0.25
Echo Tone
4500 Hz
Echo Low Cut
200 Hz
Wow
0.50 ms
Flutter
0.05 ms
Echo Drive
4.00 dB
Echo Return
-6.00 dB
SPRING
Return
-9.00 dB
Decay
2.00 s
PLATE
Decay
4.20 s
Damping
0.40
Pre-Delay
20.0 ms
Low Cut
250 Hz
Return
3.00 dB
SENDS
Ping to Echo
0.12
Hats to Echo
0.00
Perc to Echo
0.00
Point at a control: what it is and what it does is told here.



Dub: the space.



Dub: the grain cloud.

The dub chain: a tape echo with wow, flutter and drive, a spring, a plate. The ping, the chord and the kit's buses send into it, and the composer throws single hits into the echo on the 8-bar lines. The room is the hall the hats, the percussion and the ping live in; the grain cloud hears the ping and the chord and scatters them.

DUB

Control	Key	Range	Default
Echo Time	`dub.echo_time`	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/16
Feedback	`dub.feedback`	0 .. 1.2	0.25
Echo Tone	`dub.tone`	800 .. 12000 Hz	4500 Hz
Echo Low Cut	`dub.low_cut`	40 .. 800 Hz	200 Hz
Wow	`dub.wow`	0 .. 3 ms	0.5 ms
Flutter	`dub.flutter`	0 .. 0.5 ms	0.05 ms
Echo Drive	`dub.drive`	0 .. 18 dB	4 dB
Echo Return	`dub.echo_return`	-60 .. 6 dB	-3 dB
Spring Return	`dub.spring`	-60 .. 6 dB	-9 dB
Spring Decay	`dub.spring_decay`	0.3 .. 6 s	2 s
Plate Decay	`dub.plate_decay`	0.5 .. 12 s	4.2 s
Plate Damping	`dub.plate_damping`	0 .. 1	0.4
Plate Pre-Delay	`dub.plate_predelay`	0 .. 100 ms	20 ms
Plate Low Cut	`dub.plate_low_cut`	60 .. 800 Hz	250 Hz
Plate Return	`dub.plate_return`	-60 .. 18 dB	6 dB
Ping to Echo	`dub.ping_send`	0 .. 1	0.12
Hats to Echo	`dub.hats_send`	0 .. 1	0
Perc to Echo	`dub.perc_send`	0 .. 1	0

SPACE

Control	Key	Range	Default
Return	`space.level`	-60 .. 12 dB	0 dB
Size	`space.size`	0.3 .. 3	0.9
Decay	`space.decay`	0.2 .. 12 s	1.2 s
Damping	`space.damping`	0 .. 1	0.5
Pre-Delay	`space.predelay`	0 .. 100 ms	15 ms
Low Cut	`space.low_cut`	60 .. 800 Hz	250 Hz
High Cut	`space.high_cut`	1000 .. 18000 Hz	6000 Hz
Hats Send	`space.hats_send`	0 .. 1	0.25
Perc Send	`space.perc_send`	0 .. 1	0.3
Ping Send	`space.ping_send`	0 .. 1	0.4

CLOUD

Control	Key	Range	Default

Level	`cloud.level`	-60 .. 6 dB	-60 dB
Density	`cloud.density`	0.5 .. 60 /s	12 /s
Grain Size	`cloud.size`	20 .. 1000 ms	180 ms
Pitch	`cloud.pitch`	0 .. 1	0.3
Spray	`cloud.spray`	0.05 .. 4.5 s	2 s
From Ping	`cloud.ping_send`	0 .. 1	0.8
From Chord	`cloud.chord_send`	0 .. 1	0.6
Plate Send	`cloud.plate_send`	0 .. 1	0.5

Mixer and Master





The Mixer tab's second page: the buses and the master.



The Mixer tab's third page: the decks.

The Mixer tab has three pages. Console: a channel strip per part -- kick, rumble, sub, the kit's lanes, the tonal voices, the dub -- with its level, its mute, its sound and its meter, and the output with its loudness at the end (a narrow window scrolls the console sideways). Buses and Master: the parameters below. Decks: the DJ mixer's channels.

The meters: each deck after its mixer channel, the output, and the loudness of the last 400 ms. Each deck's track bus has a group high pass (the form's filter sweeps), a tilt of the highs against the lows, and the glue -- a bus compressor mixed in at 35 per cent -- after which the loudness correction sits. The master keeps the side under Mono Below mono, sets the level, has an optional vinyl cut, a clipper at four times the rate and a true-peak limiter.

The motion is the free-running modulation of the whole instrument: the hats' cutoff and level, their decay, the percussion's cutoff, the rumble's drive, slow LFOs of prime periods that never line up.

A deck's channel is a DJ mixer's: the fader, a three-band isolator (200 Hz and 2.5 kHz, Linkwitz-Riley, each band from a kill to +6 dB), a filter that is a low pass to the left and a high pass to the right, and the send into the mixer's effects. In a set the composer moves them as a DJ would; in the plugin they are yours too.

MIX

Control	Key	Range	Default
Hats Level	`mix.hats_level`	-24 .. 12 dB	-3.5 dB
Hats Cut	`mix.hats_cut`	200 .. 20000 Hz	20000 Hz

Perc Level	`mix.perc_level`	-24 .. 12 dB	-6.5 dB
Perc Cut	`mix.perc_cut`	200 .. 20000 Hz	20000 Hz
Drum Saturation	`mix.drum_sat`	0 .. 1	0.35
Duck Low	`mix.duck_low`	0 .. 18 dB	10 dB
Duck Mid	`mix.duck_mid`	0 .. 12 dB	3 dB
Group Low Cut	`mix.low_cut`	20 .. 500 Hz	20 Hz

MASTER

Control	Key	Range	Default
Level	`master.level`	-24 .. 12 dB	0 dB
Glue Threshold	`master.threshold`	-40 .. 0 dB	-16 dB
Glue Ratio	`master.ratio`	1 .. 10	2
Clip Drive	`master.clip`	0 .. 9 dB	2 dB
Ceiling	`master.ceiling`	-6 .. 0 dBTP	-1 dBTP
Mono Below	`master.mono_below`	40 .. 250 Hz	120 Hz
Tilt	`master.tilt`	-12 .. 12 dB	10 dB
Vinyl Cut	`master.cut`	off / on	off

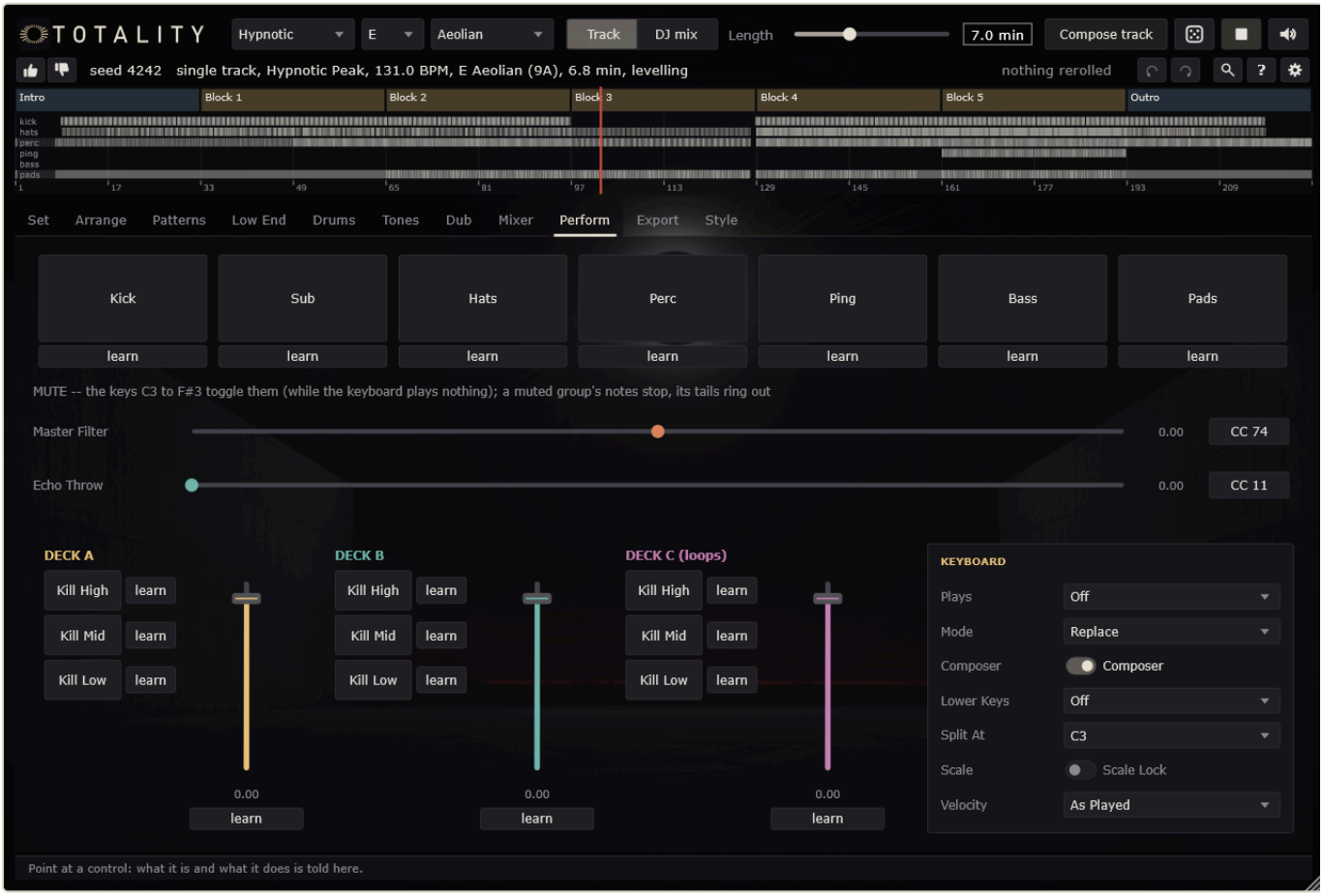
MOTION

Control	Key	Range	Default
Amount	`motion.amount`	0 .. 1	1
Hats Cut	`motion.hats_cut`	0 .. 1.5 oct	0.4 oct
Hats Level	`motion.hats_level`	0 .. 6 dB	1.5 dB
Hat Decay	`motion.hat_decay`	0 .. 0.5	0.2
Perc Cut	`motion.perc_cut`	0 .. 1.5 oct	0.4 oct
Rumble Drive	`motion.rumble_drive`	0 .. 6 dB	2 dB

DECK (3 INSTANCES)

Control	Key	Range	Default
Fader	`deckN.fader`	-60 .. 6 dB	0 dB
Low	`deckN.low`	-60 .. 6 dB	0 dB
Mid	`deckN.mid`	-60 .. 6 dB	0 dB
High	`deckN.high`	-60 .. 6 dB	0 dB
Filter	`deckN.filter`	-1 .. 1	0
FX Send	`deckN.fx_send`	0 .. 1	0

Perform



As at a mixer. Seven mutes: a muted group's notes stop and its tails ring out -- the kick out, the classic move. The master filter takes the whole mix, the echo throw sends it into the mixer's tape echo. Each deck has its kills and its fader. Every control can be learned from a MIDI controller: click learn, move the controller (a right click forgets it). By default controller 74 (brightness) moves the filter and the expression pedal (11) the throw -- the same in every generator of the family --, and the keys from middle C up (C to F#) toggle the mutes of kick, sub, hats, perc, ping, bass and pads. The bindings are saved with the project. While a headset sends its hands, a box beside the mutes shows what they do (The panel).

The keyboard (the Keyboard group): Keyboard Plays sends the keys of a MIDI keyboard to a voice instead of the mutes -- the kit (C1 the kick, C#1 to C2 the twelve lanes), the bass, the 303, the ping, the chord, the drone, or by channel (1 kit, 2 bass, 3 303, 4 ping, 5 chord, 6 drone, 10 kit) -- with the sound its page has, past the mutes. Replace leaves that voice's generated notes out, Layer plays over them; Composer off leaves every generated note out, and only what is played sounds, through the mix and the effects as composed. The keys play while the music runs, each on its sample; an export plays what was composed.

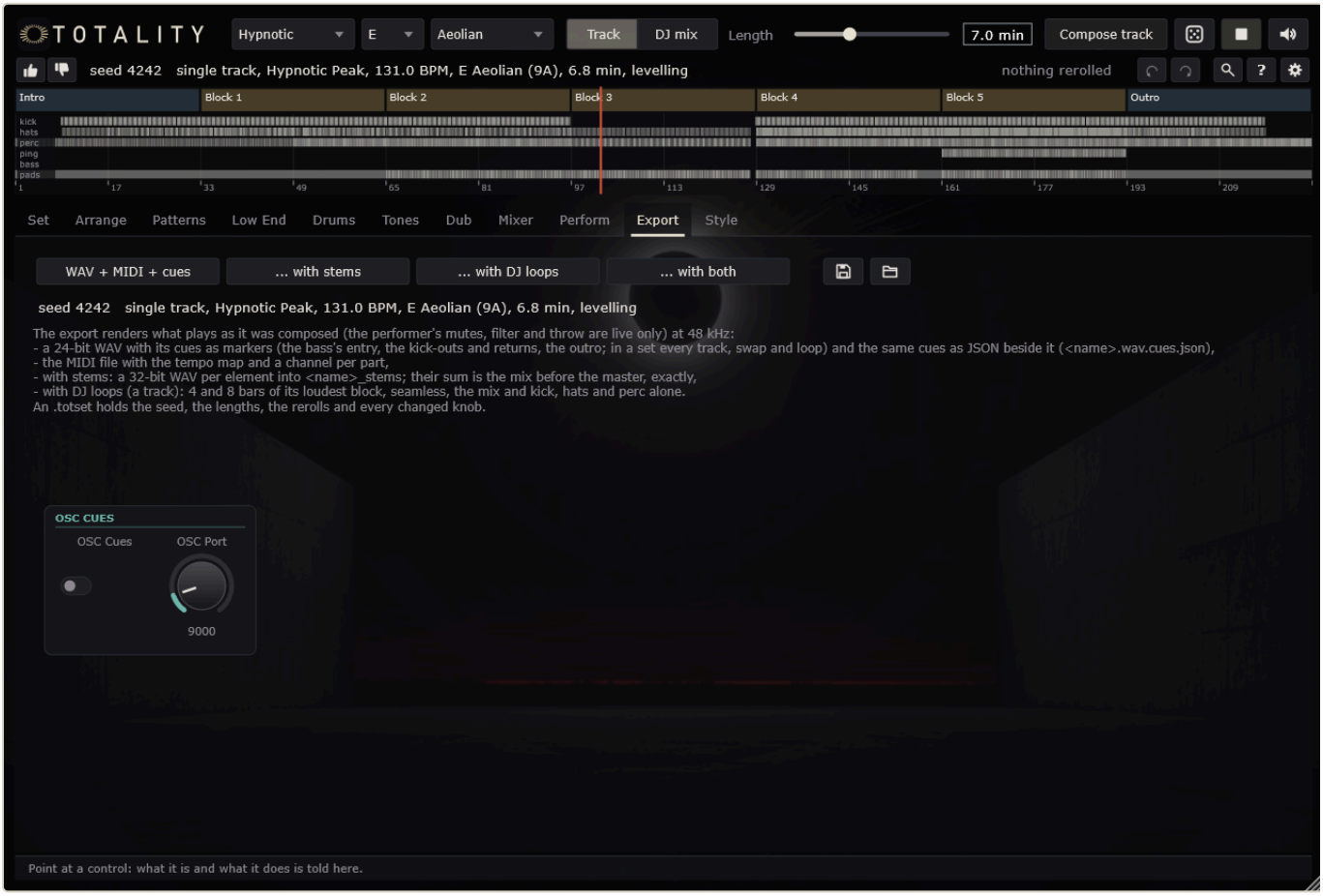
Three options of the Keyboard group, each off until you choose it: Lower Keys Play splits the keyboard -- the keys below Split At (C1 to C5, C3 to begin with) play the voice chosen there, the keys from it upwards the voice of Keyboard Plays; Scale Lock moves a key outside the track's key to the nearest note inside it, so no wrong note can sound; Velocity Curve shapes the touch -- As Played, Soft (a light touch is enough), Hard (it takes a firm one) or Fixed (every note at 100, for pads that send no velocity). A key's release always ends the note its press started, even when an option changed while it was held.

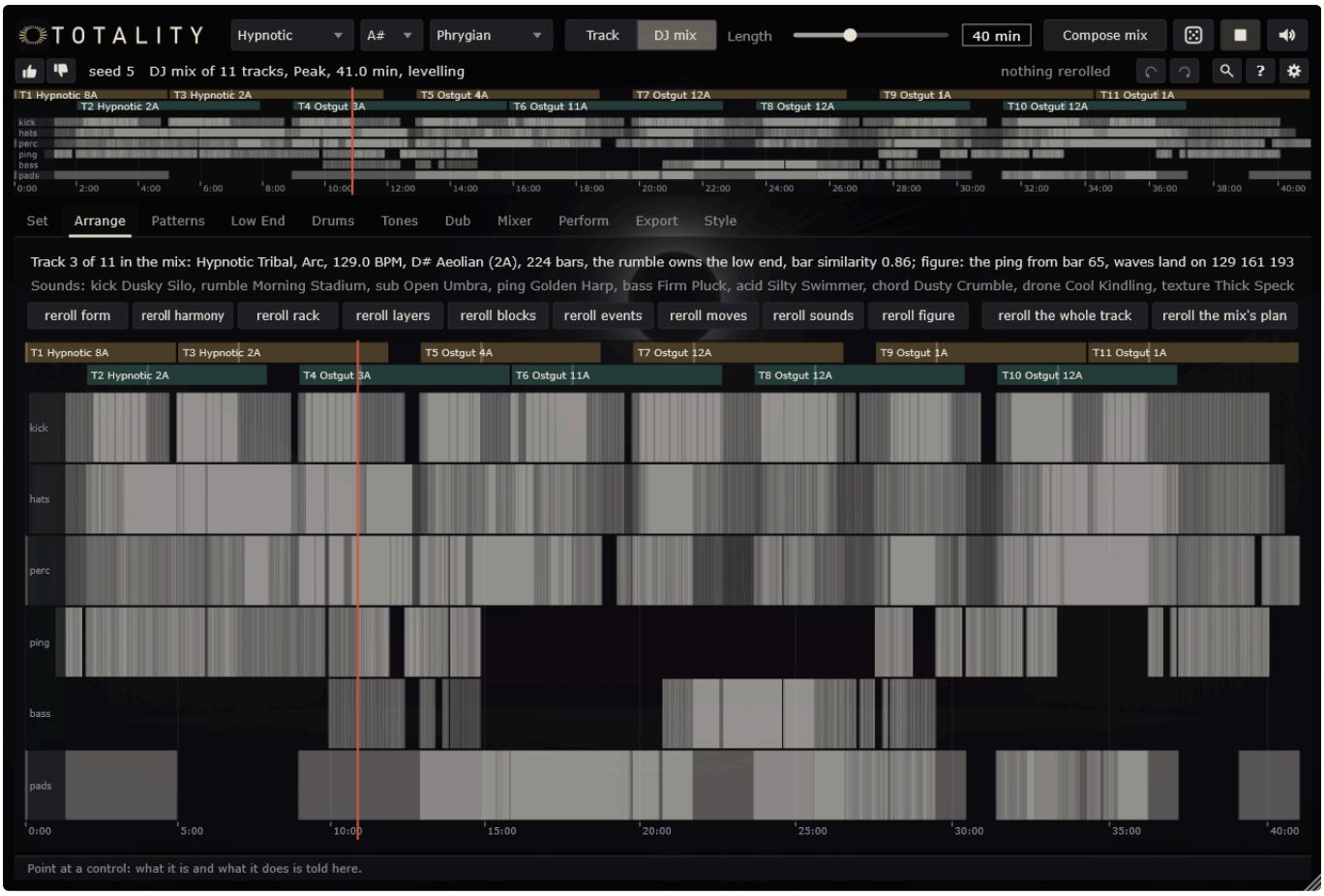
The perform controls are live only: an export or tot_render plays the track as it was composed.

PERFORM

Control	Key	Range	Default
Master Filter	`perform.filter`	-1 .. 1	0
Echo Throw	`perform.throw`	0 .. 1	0
Mute Kick	`perform.mute_kick`	off / on	off
Mute Sub	`perform.mute_sub`	off / on	off
Mute Hats	`perform.mute_hats`	off / on	off
Mute Perc	`perform.mute_perc`	off / on	off
Mute Ping	`perform.mute_ping`	off / on	off
Mute Bass	`perform.mute_bass`	off / on	off
Mute Pads	`perform.mute_pads`	off / on	off
Keyboard Plays	`perform.keyboard_part`	Off, Kit, Bass, 303, Ping, Chord, Drone, By channel	Off
Keyboard Mode	`perform.keyboard_mode`	Replace, Layer	Replace
Composer	`perform.composer`	off / on	on
Lower Keys Play	`perform.keyboard_lower`	Off, Kit, Bass, 303, Ping, Chord, Drone	Off
Split At	`perform.keyboard_split`	C1, C2, C3, C4, C5	C3
Scale Lock	`perform.keyboard_scale`	off / on	off
Velocity Curve	`perform.keyboard_velocity`	As Played, Soft, Hard, Fixed	As Played

Export and cues





A DJ mix of 40 minutes on the Arrange page: its eleven tracks on two decks, their blends, the swaps, a ruler of minutes.

The export renders what plays at 48 kHz: a 24-bit WAV with cue markers (the bass's entry, the kick-outs and returns, the outro; in a set every track, swap and loop) and the same cues as JSON beside it, for DJ software that does not read the WAV's markers, and a rekordbox collection beside it (<name>.wav.rekordbox.xml: the beat grid, following a set's tempo ramps bar by bar, every cue a memory cue and the first eight hot cues; import it in rekordbox under Preferences > Advanced > rekordbox xml); the MIDI file with the tempo map and a channel per part; on request the stems -- a WAV per element, whose sum is exactly the mix before the master -- and the DJ loops of a track: 4 and 8 bars of its loudest block, seamless, as the mix and as kick, hats and percussion alone. An .totset file holds the seed, the lengths, the rerolls and every changed knob.

The OSC cues tell a visualiser such as Kaleidoscope what the score knows, at the moment it is heard: the beat and the tempo, the bar, the block's name, the form's operation, the key. They go to this machine at the port given, or to TOT_CUE_HOST.

```
/tot/beat i f      beat number, tempo
/tot/bar i         bar number
/tot/block s       the block's name ("Intro", "Block 3", "Reduction", a set's track)
/tot/op s          the operation ("add ride", "kick out")
/tot/key s         the key as a Camelot label ("8A")
```

CUE

Control	Key	Range	Default
OSC Cues	`cue.enabled`	off / on	off
OSC Port	`cue.port`	1024 .. 65535	9000

Style



The numbers of the profile the knobs describe, and the references' measurements per style that the profiles are fitted to. A style of your own: Take the profile's numbers, change what you like, switch Use on -- the tempo range, the weights of the three forms, the length, the rack's mutation and reroll, polymeters, fills, swing, the filtered edges, the longest kick-out, the events, the corridor of bar similarity the composer's candidates are chosen against, and the loudness the Leveler brings each track to.

CUSTOM

Control	Key	Range	Default
Use Custom Style	`custom.use`	off / on	off
Tempo Low	`custom.bpm_low`	120 .. 140 BPM	128 BPM
Tempo High	`custom.bpm_high`	120 .. 140 BPM	133 BPM
Arc Weight	`custom.arc`	0 .. 1	0.5
Peak Weight	`custom.peak`	0 .. 1	0.2
Endless Weight	`custom.endless`	0 .. 1	0.3
Sub Owns Low	`custom.sub_chance`	0 .. 1	0.3
Blocks Low	`custom.blocks_low`	4 .. 12	6
Blocks High	`custom.blocks_high`	4 .. 12	8
Mutation	`custom.mutation`	0 .. 1	0.25

Reroll	`custom.reroll`	0 .. 1	0.15
Polymeter	`custom.polymeter`	0 .. 1	0.8
Fills	`custom.fill`	0 .. 1	0.15
Filtered Edges	`custom.edge`	0 .. 1	0.5
Longest Kick-out	`custom.max_reduction`	4 .. 32 bars	32 bars
Swing Low	`custom.swing_low`	50 .. 62 %	51 %
Swing High	`custom.swing_high`	50 .. 62 %	56 %
Events	`custom.event_rate`	0 .. 1	0.35
Throw Share	`custom.throw_share`	0 .. 1	0.5
Density Cap	`custom.density_cap`	3 .. 12	9
Bar Similarity	`custom.similarity`	0.7 .. 1	0.9
Target LUFS	`custom.peak_lufs`	-14 .. -7 LUFS	-10 LUFS

With a DAW and other apps

Four things connect Totality to the software around it and to the rest of the family (02.10.2026). The outputs, Link and the jam are off until they are wanted, the MIDI goes nowhere until a track takes it, and only the jam -- as a follower -- changes what the music sounds like.

MIDI out: in a DAW the plugin also sends what it plays -- every note of every part at the moment it sounds, a channel per part as in the MIDI file of an export -- and leaves the muted parts out. Record it on a MIDI track to keep a part as notes, or route it to another instrument to double it. A stop or a jump of the transport sends all notes off.

Outputs of their own: besides the main output the plugin has a stereo output for each of the stems -- the kick, the rumble, the sub, the hats, the percussion, the ping, the room, the bass, the 303, the chord, the drone, the texture, the dub, the cloud and the DJ effects. They are off to begin with; switched on in the DAW (in most hosts: the plugin's output configuration, or its multi-output version), each carries its part before the master, while the main output plays on as before.

Ableton Link (the standalone; Settings > Ableton Link): with another Link app on the network -- Live, Bitwig, a phone app, another generator of the family -- Totality takes the session's tempo, lines its bars up with the session's and starts and stops with it; alone in a session it offers its own tempo to whoever joins. In a DAW the host's transport leads, as before.

The family jam (Settings > Family jam, in the plugin and in the standalone; off to begin with): the five instruments of the family -- Totality, Parhelion, Ephemeris, Phosphene and Noctuary -- play as one band on the local network. One leads: its key, the energy of its sections, its breaks and its drops go out to the others. Those set to Follow take them up -- in Totality: the sub, the ping, the bass, the 303, the chord, the drone and the texture move to the leader's key while the drums play as written; its filter closes a little with a quiet section of the leader's (while the Perform filter stands in the middle); and in the leader's breaks the kick, the sub, the bass and the 303 are out until its drop. A new root comes in at the follower's next bar line, by the shortest way, the mode with its next track. Tempo and bars come from Ableton Link in the standalone (switch it on in each of them) or from the DAW's transport, so that a section the leader announces lands on the same bar everywhere. As the leader Totality says its track's key and, a beat ahead, each block as a section -- an intro, a groove, a build, a drop, a break, an outro --, with its energy. The jam stays on this network (UDP multicast to 239.255.77.77, port 9177, not past the router), instruments on one machine find each other as well, and a follower that has not heard its leader for four seconds goes back to its own key and its own form. The last line of the submenu says what it does: off, leading, or whom it follows.

On a Mac: every release has a build for Apple Silicon (macOS 12 or newer), the standalone, the VST3 and the Audio Unit (Logic, GarageBand, MainStage; passed by Apple's auval), built and tested on GitHub's machines. It is signed ad hoc, not notarized: macOS asks once (README-macOS.txt in the zip says how), and it has not yet been played on a real Mac -- reports are welcome.

On Linux: every release has an archive for x86-64 (glibc 2.35 or newer: Ubuntu 22.04, Debian 12, Fedora 36 and later) with the standalone, the VST3, the LV2 (Ardour, Carla, Reaper, Qtractor) and the renderer, built and tested on GitHub's Ubuntu machines and tried under WSL. README-Linux.txt in it says where everything goes: the VST3 into ~/.vst3, the LV2 into ~/.lv2, the standalone started where it lies, sound through ALSA or JACK (PipeWire serves both).

Demos: the release "demos" holds a track per style as an MP3 and one of them as a video, with pictures by KaleidoscopeEnhanced cut to the track's own score cues; Tools/demo/make_demos.py makes them all again (tot_render).

KaleidoscopeEnhanced understands the score cues of all five instruments as they come (since 02.10.2026) -- Totality's blocks and keys, Parhelion's sections, Ephemeris' phases, Phosphene's sections and drops, Noctuary's bars, keys and scenes -- and cuts its pictures to them, on a drop at once.

Meta Quest

The whole generator runs on the headset (bin\quest\TotalityQuest.apk, see Quest/README.md). The hands are the hands at the mixer: left pinch play and stop, right pinch kick out and in, both hands together the next track; the left hand's height the master filter, the right's the echo throw -- the grammar of every generator's headset (The panel). Above and ahead of the player the Eclipse turns at the size of a room. A file tot.cfg in the app's folder sets the seed, the style, a set's length and any knob; bridge_host (and bridge_port, 9103) send the hands to Totality on a computer instead, which then plays with them as with its own; audio=0 leaves the headset silent.

The command line

tot_render composes and renders without a window, the same samples the plugin plays:

```
tot_render --seed 7 --out track.wav --midi track.mid --stems stems --loops loops
tot_render --seed 2026 --set-minutes 120 --out set.wav --score-json set.json
tot_render --set "compose.style=Dub; compose.minutes=8" --seed 3 --out dub.wav
tot_render --list                  every parameter with its value
tot_render --bench --quality quest how fast, at the headset's quality
```

The evaluation (Tools/eval_report.py) measures rendered tracks as the references were measured and writes a report of what lies outside their corridor.